

R Series: Data Sets

In this eighteenth article in the R series, we take a look at the various data sets available in R.

The data sets available in R cover a wide range of fields. We have data sets that give information on the performance of automobiles, the approval rating of the presidents of the United States, the magnitude of the earthquakes around Fiji, the number of international passengers travelling in aeroplanes over a certain period, and much more.

mtcars

We have been using the *mtcars* data set which contains information on automobiles, fuel consumption and performance from the 1974 *Motor Trend* US magazine. The data frame has 32 entries and has 11 numeric fields, as shown below:

```
> head(mtcars)
      mpg cyl disp  hp drat   wt  qsec vs am gear carb
Mazda RX4  21.0   6  160 110 3.90 2.620 16.46  0   1   4   4
Mazda RX4 Wag 21.0   6  160 110 3.90 2.875 17.02  0   1   4   4
Datsun 710   22.8   4  108  93 3.85 2.320 18.61  1   1   4   1
Hornet 4 Drive 21.4   6  258 110 3.08 3.215 19.44  1   0   3   1
Hornet Sportabout 18.7  8 360 175 3.15 3.440 17.02  0   0   3   2
Valiant     18.1   6  225 105 2.76 3.460 20.22  1   0   3   1
```

The various fields are described below:

Field	Description
mpg	Miles/gallon
cyl	Number of cylinders
disp	Displacement
hp	Horse power
drat	Rear axle ratio
wt	Weight in 1000 lbs
qsec	1/4 mile time
vs	Engine (0=V-shaped, 1=straight)
am	Transmission (0=automatic, 1>manual)
gear	Number of gears
carb	Number of carburetors

airquality

The *airquality* data set provides air quality measurements in New York between May and September 1973. The data was obtained by the New York State Department of Conservation and the National Weather Service. The data frame has 153 observations and six variables, as follows:

```
> head(airquality)
      Ozone Solar.R Wind Temp Month Day
1      41     190   7.4   67     5    1
2      36     118   8.0   72     5    2
3      12     149  12.6   74     5    3
4      18     313  11.5   62     5    4
5      NA      NA  14.3   56     5    5
6      28      NA  14.9   66     5    6
```

The description of the numeric fields is as follows:

Field	Description
Ozone	Ozone in parts per billion
Solar.R	Solar radiation in Langleys
Wind	In mph
Temp	Temperature in degrees Fahrenheit
Month	Values: 1-12
Day	Values: 1-31

AirPassengers

An example of time series data is provided by the *AirPassengers* data set given by Box & Jenkins. It contains the monthly totals of international airline passengers (in thousands) between 1949 and 1960. The data set is illustrated below for reference:

```
> AirPassengers
      Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
1949 112 118 132 129 121 135 148 148 136 119 104 118
1950 115 126 141 135 125 149 170 170 158 133 114 140
1951 145 150 178 163 172 178 199 199 184 162 146 166
```

...

presidents

Another example of time series data is the approval rating for the President of the United States for various quarters from 1945 till 1974 in the *presidents* data set. It has 120 values and has been provided by the Gallup Organisation.